Predicting Strategic Selection

Wednesday, September 2 · Building the State in Real Time, Week 7

All teams · today's working task on the strategic-selection problem

The task

The government's inclusion codes reveal that 19 of the 61 selected municipalities qualify on none of the three criteria at the reconstructed cutoffs. These 19 are the strategic-selection residual, and today's job is to evaluate which factors best predict that selection. From Canvas, download the pool file: `w7_strategic19_zcomp.csv`. One row per municipality: the 19 strategic cases (flag `strategic_target`) and 167 comparison municipalities, not selected and qualifying on nothing, each one assigned to the window of its nearest strategic case.

The design: window-based sampling with stratum fixed effects

Each strategic case is compared only to non-selected municipalities ranked within ±10 positions of it on the z-composite index (`rank_final`; zero-insertion perception rule). The `stratum` column assigns every control exclusively to its nearest case: 19 strata, one case each, 3–21 controls (median 8). All estimation includes stratum (window) fixed effects, so every comparison is *within-window*: a factor predicts strategic selection only if it distinguishes the chosen municipality from its own rank-neighbors, never from municipalities in a different part of the risk distribution. This is a matched-set design; treat it exactly as you would pair fixed effects.

Candidate predictors and where to find them

Aim for two or three predictors per specification; nineteen cases cannot support more. Candidates, with sources:

- **Federal-corridor position**: SICT/IMT Red Nacional de Caminos shapefile; corridor membership for the Balam and Cero Robos priority routes (buffer-intersect; see the highway trade-risk script, `hsif_week7_highway_trade_risk.R`)
- **Cargo theft (robo a transportista), 2025 level and change**: SESNSP Incidencia Delictiva Municipal on `datos.gob.mx`; the same file as the criteria counts
- **Metropolitan-zone membership and metro name**: CONAPO/INEGI/SEDATU delimitation of metropolitan zones 2020
- **State capital; adjacency to a capital**: Marco Geoestadístico catalog
- **Tourism designation**: SECTUR pueblos mágicos and DataTur destination lists
- **Airports and ports**: AFAC airport catalog; ASIPONA port list
- **Economic complexity (ECI)**: Data México API, or the reconstruction script `hsif_week7_municipal_eci.py`

- **Conflict salience**: ACLED Mexico events 2024–2025 (registration); RNPDNO disappearances; the 2024 electoral-violence trackers
- **Public expectation**: `public_search_list_51.csv` (Canvas): pre-announcement press list; on-list-but-not-qualifying is a salience signal
- **Political alignment**: INE 2024 municipal results; state-level alignment with the federal government

Several of these overlap with the domain tables other teams are assembling (corridors and airports with Team 4, ECI and cargo flows with Team 1, metro structure with Team 2). **Before pulling anything, check the group chat and coordinate**: claim variables there, share cleaned CVEGEO-keyed files, and do not duplicate a pull another team has finished.

Approaches

- **OLS (linear probability) with stratum fixed effects**: `feols(strategic_target ~ x1 + x2 | stratum)` with standard errors clustered by stratum. Transparent coefficients; the baseline
- **Conditional logit**: one case per stratum makes this a within-window choice model: the probability the government picked municipality i from its window. In R: `clogit(strategic_target ~ x1 + x2 + strata(stratum))` from `survival`. The preferred specification; report it alongside OLS
- **Machine learning, used honestly**: lasso or ridge logistic on within-stratum demeaned predictors (regularization chosen by leave-one-stratum-out); tree ensembles only as an exploratory ranking of candidate predictors, never as the main approach. With 19 positives, flexible learners will memorize; the guard is the evaluation below
- **Evaluation**: leave-one-stratum-out throughout. Score each model by where it ranks the true case within its held-out window: report top-1 and top-3 counts against their chance values (chance top-1 $\approx$ 2.8 of 19 at these window sizes). This scoreboard is more honest than an AUC with nineteen positives

Sensitivity tests (run at least two)

- **Window width**: rebuild the sample pool at ± 5 and ± 10 and re-estimate; conclusions that hold only at one width are window artifacts
- **Drop the stratum fixed effects**: run the pooled specification on the same sample; the gap between pooled and within-window coefficients measures how much of a predictor's apparent power was across-window (rank-level) variation
- **Randomization inference**: within each stratum, randomly reassign the "strategic" label to one member; re-estimate; repeat 1,000 times. The share of permuted coefficients (or LOO top-1 counts) exceeding the observed one is the RI p-value. This is the cleanest statement of uncertainty this small design permits

Deliverable

By 5:00pm: a one-page results memo (predictors tested, preferred specification, LOO scoreboard, at least two sensitivity tests, and the sentence stating what "strategic importance" appears to operationalize), plus the code. Findings feed Thursday's validation exercises and the Friday briefing.